



Faculty of Computer Applications

**MCA Sem-1
(AY-2026-27)**

Lab Book

**Sub Code: 05MC0108
(DBMS)**

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1	Write a PL/SQL block that find 1. Area Of Circle 2. Area Of Rectangle 3. Circumference of Circle
2	Write a PL/SQL block to accept product name, qty and price from user and then calculate discount in Rs. based on the given (%).
3	Write a PL/SQL block to calculate the total, percentage and grade of student based on his/her Rollno from RESULT table. (Create RESULT table with Rollno, Name, Sub1, Sub2, Sub3, Sub4, Sub5, Total, Per, Grade attributes with appropriate data type).
4	1. Write a program that prints value 1 to 100 numbers using FOR LOOP. 2. Write a program that prints value 1 to 100 number using LOOP Command. 3. Write a program that prints value 1 to 100 number using WHILE LOOP Command
5	Write a PL/SQL block which displays gross salary of employees as per user input EID. (Consider EMP table with EID, EName, Deptno, Deptname Gender, Age, BasicSal) with appropriate data types.) Gross_Salary: BASICSAL + (DA + HRA + Medical) – PF. Rules: HRA = 15% of basic, DA = 50% of basic, Medical = Rs. 500, PF = 10% of basic.
6	Write a PL/SQL block which accepts measurement in feet and displays it in cm, inch and meter.
7	Write a PL/SQL block which converts temperature from Celsius to Fahrenheit.
8	Write a PL/SQL block to delete the record of employee for given EID.
9	Write a PL/SQL block that calculates the simple interest based on given principal amount, rate of interest and number of years.
10	Write a PL/SQL block to calculate the square and cube of the given number.

11	Write a simple procedure that increases by the salary of employees for the given department no. by percentage inputted by the user using IN parameter. Also handle the exception if inputted department number not found.
12	Write a procedure that search whether the given employee number is present or not in the table if present then print name, salary and department number of that employee. (Use both IN and OUT mode variables) and also Write a PL/SQL block to call the SEARCH_EMP procedure.
13	Write a procedure that accept employee ID and remove records from employee table. Also create a calling program to call procedure.
14	Write a function that returns the age, category of an employee. Categories: ● Age < 18 → Minor ● 18–60 → Adult ● 60 → Senior
15	Write a PACKAGE which includes Package Declaration, Package Body and call the individual object from the Package.

1. Write a PL/SQL block that find:

1. Area of Circle
 2. Area of Rectangle
 3. Circumference of Circle
-

```
/*
    Write a PL/SQL block that find
    1.Area of circle(pi*radius*radius)
    2.Area of Rectangle(length*width)
    3.Circumference of circle(2*PI*RADIUS)
*/

ACCEPT CHOICE PROMPT "Enter Your Choice: 1.Area of circle ,2.Area of Rectangle
,3.Circumference of circle "
SET SERVEROUTPUT ON
SET VERIFY OFF
SET FEEDBACK OFF

DECLARE
    CHOICE NUMBER:=&CHOICE;
    PI NUMBER(7,2):=3.14;
    RADIUS NUMBER(5);
    LENGTH NUMBER(5);
    WIDTH NUMBER(5);
BEGIN
    --CHOICE:=&CHOICE;

    DBMS_OUTPUT.PUT_LINE(' ');
    IF (CHOICE=1) THEN
        RADIUS:=&RADIUS;
        DBMS_OUTPUT.PUT_LINE('Area of circle is : ' || PI*(POWER(RADIUS,2)));
    ELSIF (CHOICE=2) THEN
        LENGTH:=&LENGTH;
        WIDTH:=&WIDTH;
        DBMS_OUTPUT.PUT_LINE('Area of Rectangle is : ' || LENGTH*WIDTH);
    ELSIF (CHOICE=3) THEN
        DBMS_OUTPUT.PUT_LINE('Circumference of circle is : ' || 2*PI*RADIUS);
    ELSE
        DBMS_OUTPUT.PUT_LINE('Invalid Choice');
```

```
END IF;  
END;  
/  
SET SERVEROUTPUT OFF  
SET VERIFY ON  
SET FEEDBACK ON
```

Output:

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p1.sql  
Enter Your Choice: 1.Area of circle ,2.Area of Rectangle ,3.Circumference of circle :- 1  
Enter value for radius: 12  
Enter value for length: 8  
Enter value for width: 5  
  
Area of circle is : 452.16
```

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p1.sql  
Enter Your Choice: 1.Area of circle ,2.Area of Rectangle ,3.Circumference of circle :- 2  
Enter value for radius: 12  
Enter value for length: 45  
Enter value for width: 40  
  
Area of Rectangle is : 1800
```

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p1.sql  
Enter Your Choice: 1.Area of circle ,2.Area of Rectangle ,3.Circumference of circle :- 3  
Enter value for radius: 10  
Enter value for length: 40  
Enter value for width: 45  
  
Circumference of circle is : 62.8
```

2. Write a PL/SQL block to accept product name, qty and price from user and then calculate discount in Rs. based on the given (%).

```
/*
Write a PL/SQL block to accept product name ,qty and price from user
and then calculate discount in Rs.based on the given (%)
*/

SET SERVEROUTPUT ON
SET VERIFY OFF
SET FEEDBACK OFF

ACCEPT PN PROMPT "Enter product name :- "
DECLARE
    P_NAME      VARCHAR(20);
    QTY    NUMBER(5);
    PRICE  NUMBER(7,2);
    TP     NUMBER(7,2);
    DISCOUNT  NUMBER(5);
    FP     NUMBER(7,2);
BEGIN
    P_NAME:='&PN';
    QTY:=&QTY;
    PRICE:=&PRICE;
    DISCOUNT:=&DISCOUNT;

    TP:=PRICE*QTY;
    FP:=TP-(TP*DISCOUNT/100);
    DBMS_OUTPUT.PUT_LINE(' ');
    DBMS_OUTPUT.PUT_LINE('Product name is := ' || P_NAME);
    DBMS_OUTPUT.PUT_LINE('Total qty is := ' || QTY);
    DBMS_OUTPUT.PUT_LINE('Price is := ' || PRICE);
    DBMS_OUTPUT.PUT_LINE('Total Price is := ' || TP);
    DBMS_OUTPUT.PUT_LINE('Discount is := ' || DISCOUNT);
    DBMS_OUTPUT.PUT_LINE('Final Price is := ' || FP);

END;
/
SET SERVEROUTPUT OFF
```

SET VERIFY ON
SET FEEDBACK ON

Output:

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p2.sql
Enter product name :- key-board
Enter value for qty: 3
Enter value for price: 200
Enter value for discount: 12

Product name is := key-board
Total qty is := 3
Price is := 200
Total Price is := 600
Discount is := 12
Final Price is := 528
```

3. Write a PL/SQL block to calculate the total, percentage and grade of student based on his/her Rollno from RESULT table. (Create RESULT table with Rollno, Name, Sub1, Sub2, Sub3, Sub4, Sub5, Total, Per, Grade attributes with appropriate data type).

```
/*  
Write a PL/SQL block to calculate the total, percentage and grade of  
student based on his/her Rollno from RESULT table. (Create RESULT  
table with Rollno, Name, Sub1, Sub2, Sub3, Sub4, Sub5, Total, Per,  
Grade attributes with appropriate data type).  
*/
```

```
SET SERVEROUTPUT ON  
SET VERIFY OFF  
SET FEEDBACK OFF  
DECLARE  
    ROLL_NO NUMBER(5);  
    NAME VARCHAR2(30);  
    SUB1 NUMBER(3);  
    SUB2 NUMBER(3);  
    SUB3 NUMBER(3);  
    SUB4 NUMBER(3);  
    SUB5 NUMBER(3);  
    TOTAL NUMBER(5);  
    PER NUMBER(7,2);  
    GRADE VARCHAR2(30);  
BEGIN  
    ROLL_NO:=&ROLL_NO;  
    NAME:='&NAME';  
    SUB1:=&SUB1;  
    SUB2 :=&SUB2 ;  
    SUB3 :=&SUB3 ;  
    SUB4:=&SUB4;  
    SUB5:=&SUB5;  
    TOTAL:=SUB1+SUB2+SUB3+SUB4+SUB5;  
    PER:=TOTAL/5;  
  
    IF (PER>=75) THEN  
        GRADE:='O';  
    ELSIF (PER<=75)AND(PER>=70) THEN  
        GRADE:='A';
```



```
ELSIF (PER<=60)AND(PER>=70) THEN
    GRADE:='B';
ELSIF (PER<=50)AND(PER>=60) THEN
    GRADE:='C';
ELSE
    GRADE:='PASS CLASS';
END IF;
```

```
INSERT INTO
RESULT(ROLLNO,NAME,SUB1,SUB2,SUB3,SUB4,SUB5,TOTAL,PER,GRADE)VALUES(RO
LL_NO,NAME,SUB1,SUB2,SUB3,SUB4,SUB5,TOTAL,PER,GRADE);
```

```
DBMS_OUTPUT.PUT_LINE(' ');
DBMS_OUTPUT.PUT_LINE('Roll no:-' || ROLL_NO);
DBMS_OUTPUT.PUT_LINE('Name:-' || NAME);
DBMS_OUTPUT.PUT_LINE(' ');
DBMS_OUTPUT.PUT_LINE('-----Result-----');
DBMS_OUTPUT.PUT_LINE('Marks of sub1:-' || SUB1);
DBMS_OUTPUT.PUT_LINE('Marks of sub2:-' || SUB2);
DBMS_OUTPUT.PUT_LINE('Marks of sub3:-' || SUB3);
DBMS_OUTPUT.PUT_LINE('Marks of sub4:-' || SUB4);
DBMS_OUTPUT.PUT_LINE('Marks of sub5:-' || SUB5);

DBMS_OUTPUT.PUT_LINE(' ');
DBMS_OUTPUT.PUT_LINE('-----');
DBMS_OUTPUT.PUT_LINE('Your total marks : ' || TOTAL);
DBMS_OUTPUT.PUT_LINE('Percentage : ' || PER || '%');
DBMS_OUTPUT.PUT_LINE('Grade:' || GRADE);
```

```
END;
```

```
/
```

```
SET SERVEROUTPUT OFF
SET VERIFY ON
SET FEEDBACK ON
```

Output:

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p3.sql
Enter value for roll_no: 17
Enter value for name: kunjal
Enter value for sub1: 80
Enter value for sub2: 90
Enter value for sub3: 88
Enter value for sub4: 96
Enter value for sub5: 95
```

```
Roll no:-17
Name:-kunjal
```

```
-----Result-----
```

```
Marks of sub1:-80
Marks of sub2:-90
Marks of sub3:-88
Marks of sub4:-96
Marks of sub5:-95
```

```
-----
Your total marks : 449
Percentage :89.8%
Grade:0
```

4. 1. Write a program that prints value 1 to 100 numbers using FOR LOOP.
 2. Write a program that prints value 1 to 100 number using LOOP Command.
 3. Write a program that prints value 1 to 100 number using WHILE LOOP Command
-

/*

1. Write a program that prints value 1 to 100 numbers using FOR LOOP.
2. Write a program that prints value 1 to 100 number using LOOP Command.
3. Write a program that prints value 1 to 100 number using WHILE LOOP Command

*/

```
SET SERVEROUTPUT ON
SET VERIFY OFF
SET FEEDBACK OFF
```

```
DECLARE
```

```
    i NUMBER := 1;
```

```
BEGIN
```

```
    -- FOR LOOP
```

```
    DBMS_OUTPUT.PUT_LINE('--- FOR LOOP ---');
```

```
    FOR i IN 1..100 LOOP
```

```
        DBMS_OUTPUT.PUT(i || ' ');
```

```
    END LOOP;
```

```
    DBMS_OUTPUT.PUT_LINE('                ');
```

```
    -- SIMPLE LOOP
```

```
    DBMS_OUTPUT.PUT_LINE('--- SIMPLE LOOP ---');
```

```
    i := 1;
```

```
    LOOP
```

```
        DBMS_OUTPUT.PUT(i || ' ');
```

```
        i := i + 1;
```

```
        EXIT WHEN i > 100;
```

```
    END LOOP;
```

```
    DBMS_OUTPUT.PUT_LINE('');
```

```
DBMS_OUTPUT.PUT_LINE('--- WHILE LOOP ---');
i := 1;
WHILE i <= 100 LOOP
    DBMS_OUTPUT.PUT(i || ' ');
    i := i + 1;
END LOOP;
DBMS_OUTPUT.PUT_LINE('');
END;
/
SET SERVEROUTPUT OFF
SET VERIFY ON
SET FEEDBACK ON
```

Output:

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p4.sql
--- FOR LOOP ---
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30
31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57
58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84
85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100
--- SIMPLE LOOP ---
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30
31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57
58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84
85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100
--- WHILE LOOP ---
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30
31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57
58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84
85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100
```

5. Write a PL/SQL block which displays gross salary of employees as per user input EID. (Consider EMP table with EID, EName, Deptno, Deptname Gender, Age, BasicSal) with appropriate data types.) Gross_Salary: BASICSAL + (DA + HRA + Medical) – PF. Rules: HRA = 15% of basic, DA = 50% of basic, Medical = Rs. 500, PF = 10% of basic.

```
/*  
Write a PL/SQL block which displays gross salary of employees as per user input  
EID.  
Consider EMP table with EID, EName, Deptno, Deptname, Gender, Age, BasicSal  
with appropriate data types.  
Gross_Salary:= BASICSAL + (DA + HRA + Medical) – PF.  
Rules: HRA = 15% of basic, DA = 50% of basic, Medical = Rs. 500, PF = 10% of basic.  
Here for practical we have used Scott users inbuilt EMP table.  
*/
```

```
SET SERVEROUTPUT ON  
SET FEEDBACK OFF  
SET VERIFY OFF
```

```
DECLARE  
  V_EID    EMP.EID%TYPE;  
  V_NAME    EMP.ENAME%TYPE;  
  BASIC_SAL EMP.BASICSAL%TYPE;  
  DA        NUMBER;  
  HRA        NUMBER;  
  MEDICAL    NUMBER := 500;  
  PF         NUMBER;  
  GROSS_SAL  NUMBER;
```

```
BEGIN  
  V_EID := &EID;  
  
  SELECT ENAME, BASICSAL  
  INTO V_NAME, BASIC_SAL  
  FROM EMP  
  WHERE EID = V_EID;  
  
  DA := BASIC_SAL * 0.50;  
  HRA := BASIC_SAL * 0.15;
```

```

PF := BASIC_SAL * 0.10;
GROSS_SAL := BASIC_SAL + DA + HRA + MEDICAL - PF;

DBMS_OUTPUT.PUT_LINE('----- SALARY SLIP -----');
DBMS_OUTPUT.PUT_LINE('Employee ID   : ' || V_EID);
DBMS_OUTPUT.PUT_LINE('Employee Name  : ' || V_NAME);
DBMS_OUTPUT.PUT_LINE('Basic Salary   : Rs. ' || BASIC_SAL);
DBMS_OUTPUT.PUT_LINE('DA Amount      : Rs. ' || DA);
DBMS_OUTPUT.PUT_LINE('HRA Amount     : Rs. ' || HRA);
DBMS_OUTPUT.PUT_LINE('Medical Amount : Rs. ' || MEDICAL);
DBMS_OUTPUT.PUT_LINE('Provident Fund : Rs. ' || PF);
DBMS_OUTPUT.PUT_LINE('-----');
DBMS_OUTPUT.PUT_LINE('Gross Salary   : Rs. ' || GROSS_SAL);
DBMS_OUTPUT.PUT_LINE('-----');
EXCEPTION
  WHEN NO_DATA_FOUND THEN
    DBMS_OUTPUT.PUT_LINE(
      'Employee record for given EID - ' || V_EID || ' does not exist'
    );
END;
/

SET SERVEROUTPUT OFF
SET FEEDBACK ON
SET VERIFY ON

```

Output:

```

SQL> @ D:\Kunjal\DBMS\Unit_1\p5.sql
Enter value for eid: 105
----- SALARY SLIP -----
Employee ID       : 105
Employee Name     : PRIYA
Basic Salary      : Rs. 32000
DA Amount         : Rs. 16000
HRA Amount        : Rs. 4800
Medical Amount    : Rs. 500
Provident Fund     : Rs. 3200
-----
Gross Salary      : Rs. 50100
-----

```

6. Write a PL/SQL block which accepts measurement in feet and displays it in cm, inch and meter.

/*

Write a PL/SQL block which accepts measurement in feet and displays it in cm, inch and meter.

Formulas:-

1)Inches :-feet * 12

2)Centimeters :- feet * 30.48

3)Meters:-feet*0.3045

*/

SET SERVEROUTPUT ON

SET VERIFY OFF

SET FEEDBACK OFF

DECLARE

FEET NUMBER;

CM NUMBER;

INCH NUMBER;

METER NUMBER;

BEGIN

FEET := &FEET;

CM := FEET * 30.48;

INCH := FEET * 12;

METER := FEET * 0.3048;

DBMS_OUTPUT.PUT_LINE('Feet : ' || FEET);

DBMS_OUTPUT.PUT_LINE('Centimeter : ' || CM);

DBMS_OUTPUT.PUT_LINE('Inch : ' || INCH);

DBMS_OUTPUT.PUT_LINE('Meter : ' || METER);

END;

/

SET SERVEROUTPUT OFF

SET VERIFY ON

SET FEEDBACK ON

Output:

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p5.sql
Enter value for eid: 105
----- SALARY SLIP -----
Employee ID      : 105
Employee Name    : PRIYA
Basic Salary     : Rs. 32000
DA Amount        : Rs. 16000
HRA Amount       : Rs. 4800
Medical Amount   : Rs. 500
Provident Fund   : Rs. 3200
-----
Gross Salary     : Rs. 50100
-----
```

7. Write a PL/SQL block which converts temperature from Celsius to Fahrenheit.

/*

Write a PL/SQL block which converts temperature from Celsius to Fahrenheit.

Formulas:-

-Celsius to Fahrenheit :-(Celsius *9/2)+32

-Fahrenheit to Celsius :-(Fahrenheit-32)/5/9

*/

SET SERVEROUTPUT ON

SET VERIFY OFF

SET FEEDBACK OFF

ACCEPT CH PROMPT 'Enter your choice :- 1. Celsius to Fahrenheit, 2. Fahrenheit to Celsius: '

DECLARE

CELSIUS NUMBER(10,2);

FAHRENHEIT NUMBER(10,2);

CHOICE NUMBER(10);

BEGIN

CHOICE := &CH;

IF CHOICE = 1 THEN

CELSIUS := &CELSIUS;

FAHRENHEIT := (CELSIUS * 9 / 5) + 32;

DBMS_OUTPUT.PUT_LINE('Value of FAHRENHEIT :- ' || FAHRENHEIT);

ELSIF CHOICE = 2 THEN

FAHRENHEIT := &FAHRENHEIT;

CELSIUS := (FAHRENHEIT - 32) * 5 / 9;

DBMS_OUTPUT.PUT_LINE('Value of CELSIUS :- ' || CELSIUS || 'C');

ELSE

DBMS_OUTPUT.PUT_LINE('INVALID CHOICE... Enter 1 or 2');

END IF;

END;

/

Output:

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p7.sql
Enter your choice :- 1. Celsius to Fahrenheit, 2. Fahrenheit to Celsius: 1
Enter value for celsius: 12
Enter value for fahrenheit: 30
Value of FAHRENHEIT :- 53.6
```

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p7.sql
Enter your choice :- 1. Celsius to Fahrenheit, 2. Fahrenheit to Celsius: 2
Enter value for celsius: -45
Enter value for fahrenheit: 45
Value of CELSIUS :- 7.22C
SQL>
```

8. Write a PL/SQL block to delete the record of employee for given EID.

```
/*
Write a PL/SQL block to remove an employee record
for the given Employee ID.
*/

SET SERVEROUTPUT ON
SET FEEDBACK OFF
SET VERIFY OFF
DECLARE
    EMP_ID EMP.EID%TYPE;
    CNT NUMBER;
BEGIN
    EMP_ID := &EID;

    SELECT COUNT(*) INTO CNT FROM EMP WHERE EID = EMP_ID;

    IF CNT > 0 THEN
        DELETE FROM EMP WHERE EID = EMP_ID;

        COMMIT;
        DBMS_OUTPUT.PUT_LINE('Employee ID ' || EMP_ID || ' deleted successfully.'
    );
    ELSE
        DBMS_OUTPUT.PUT_LINE('Employee ID ' || EMP_ID || ' does not exist. ');
    END IF;

END;
/
SET SERVEROUTPUT OFF
SET FEEDBACK ON
SET VERIFY ON
```

Output:

```
value of EID is 105
SQL> @ D:\Kunjal\DBMS\Unit_1\p8.sql
Enter value for eid: 105
Employee ID 105 deleted successfully.
SQL>
```

9. Write a PL/SQL block that calculates the simple interest based on given principal amount, rate of interest and number of years.

/* Write a PL/SQL block to calculate Simple Interest and Total Amount. */

```
SET SERVEROUTPUT ON
SET FEEDBACK OFF
SET VERIFY OFF
DECLARE
    P NUMBER;
    R NUMBER;
    T NUMBER;
    SI NUMBER;
    AMOUNT NUMBER;
BEGIN
    P := &P;
    R := &R;
    T := &T;
    SI := (P * R * T) / 100;
    AMOUNT := P + SI;
    DBMS_OUTPUT.PUT_LINE('=====');
    DBMS_OUTPUT.PUT_LINE('Principal Amount : ' || P);
    DBMS_OUTPUT.PUT_LINE('Rate of Interest : ' || R || '%');
    DBMS_OUTPUT.PUT_LINE('Number of Years : ' || T);
    DBMS_OUTPUT.PUT_LINE('Simple Interest : ' || SI);
    DBMS_OUTPUT.PUT_LINE('Total Amount : ' || AMOUNT);
END;
/
SET SERVEROUTPUT OFF
SET FEEDBACK ON
SET VERIFY ON
```

Output:

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p9.sql
Enter value for p: 12000
Enter value for r: 10
Enter value for t: 5
=====
Principal Amount : 12000
Rate of Interest : 10%
Number of Years : 5
Simple Interest : 6000
Total Amount : 18000
```

10. Write a PL/SQL block to calculate the square and cube of the given number.

```
/*  
Write a PL/SQL block to calculate the square and cube  
of the given number.  
*/
```

```
SET SERVEROUTPUT ON  
SET FEEDBACK OFF  
SET VERIFY OFF
```

```
DECLARE  
    N NUMBER;  
    SQUARE NUMBER;  
    CUBE NUMBER;  
BEGIN  
    N := &N;  
    SQUARE := N * N;  
    CUBE := N * N * N;  
  
    DBMS_OUTPUT.PUT_LINE('Given Number : ' || N);  
    DBMS_OUTPUT.PUT_LINE('Square      : ' || SQUARE);  
    DBMS_OUTPUT.PUT_LINE('Cube       : ' || CUBE);  
  
END;  
/
```

```
SET SERVEROUTPUT OFF  
SET FEEDBACK ON  
SET VERIFY ON
```

Output:

```
cube      : 800000  
SQL> @ D:\Kunjal\DBMS\Unit_1\p10.sql  
Enter value for n: 20  
Given Number : 20  
Square      : 400  
Cube       : 8000  
SQL> _
```

11. Write a simple procedure that increases by the salary of employees for the given department no. by percentage inputted by the user using IN parameter. Also handle the exception if inputted department number not found.

```
/*
Write a simple procedure that increases by the salary of employees for the given
department no. by percentage inputted by the user using IN parameter. Also
handle the exception if inputted department number not found.
*/

SET SERVEROUTPUT ON
SET VERIFY OFF
SET FEEDBACK OFF

CREATE OR REPLACE PROCEDURE increase_salary (
    p_deptno IN NUMBER,
    p_percent IN NUMBER
)
IS
    v_count NUMBER;
BEGIN
    SELECT COUNT(*)
    INTO v_count
    FROM EMP
    WHERE DEPTNO = p_deptno;

    IF v_count = 0 THEN
        RAISE_APPLICATION_ERROR(
            -20001,
            'Department number ' || p_deptno || ' not found'
        );
    END IF;

    UPDATE EMP
    SET BASICSAL = BASICSAL + (BASICSAL * p_percent / 100)
    WHERE DEPTNO = p_deptno;

    COMMIT;

    DBMS_OUTPUT.PUT_LINE(
```

```
'Salary increased by ' || p_percent ||  
'% for department ' || p_deptno  
);  
  
EXCEPTION  
  WHEN OTHERS THEN  
    DBMS_OUTPUT.PUT_LINE('Error: ' || SQLERRM);  
END;  
/  
SET SERVEROUTPUT OFF  
SET VERIFY ON  
SET FEEDBACK ON
```

Output:

12. Write a procedure that search whether the given employee number is present or not in the table if present then print name, salary and department number of that employee. (Use both IN and OUT mode variables) and also Write a PL/SQL block to call the SEARCH_EMP procedure.

```
/*  
Write a procedure that search whether the given employee number is  
present or not in the table if present then print name, salary and  
department number of that employee. (Use both IN and OUT mode  
variables) and also Write a PL/SQL block to call the SEARCH_EMP  
procedure.  
*/
```

```
SET SERVEROUTPUT ON  
SET VERIFY OFF  
SET FEEDBACK OFF
```

```
CREATE OR REPLACE PROCEDURE SEARCH_EMP (  
    P_EID    IN NUMBER,  
    P_ENAME  OUT VARCHAR2,  
    P_SALARY OUT NUMBER,  
    P_DEPTNO OUT NUMBER  
)  
IS  
BEGIN  
    SELECT ENAME, BASICSAL, DEPTNO  
    INTO P_ENAME, P_SALARY, P_DEPTNO  
    FROM EMP  
    WHERE EID = P_EID;  
  
EXCEPTION  
    WHEN NO_DATA_FOUND THEN  
        P_ENAME := NULL;  
        P_SALARY := NULL;  
        P_DEPTNO := NULL;  
  
        DBMS_OUTPUT.PUT_LINE(  
            'Employee number ' || P_EID || ' not found.'  
        );  
END;
```


/

SET SERVEROUTPUT OFF

SET VERIFY ON

SET FEEDBACK ON

Output:

13. Write a procedure that accept employee ID and remove records from employee table. Also create a calling program to call procedure.

```
/*
Write a procedure that accept employee ID and remove records from employee
table. Also create a calling program to call procedure.
*/

SET SERVEROUTPUT ON
SET VERIFY OFF
SET FEEDBACK OFF

CREATE OR REPLACE PROCEDURE REMOVE_EMPLOYEE (
  P_EID IN EMP.EID%TYPE
)
IS
BEGIN
  DELETE FROM EMP
  WHERE EID = P_EID;

  IF SQL%ROWCOUNT = 0 THEN
    DBMS_OUTPUT.PUT_LINE(
      'Employee ID ' || P_EID || ' not found.'
    );
  ELSE
    DBMS_OUTPUT.PUT_LINE(
      'Employee ID ' || P_EID || ' deleted successfully.'
    );
  END IF;

END;
/

SET SERVEROUTPUT OFF
SET VERIFY ON
SET FEEDBACK ON
```

Output:

14. Write a function that returns the age, category of an employee.

Categories:

- Age < 18 → Minor
 - 18–60 → Adult
 - 60 → Senior
-

/*

Write a function that returns the age, category of an employee.

Categories:

- Age < 18 → Minor
- 18–60 → Adult
- 60 → Senior

*/

SET SERVEROUTPUT ON

SET VERIFY OFF

SET FEEDBACK OFF

CREATE OR REPLACE FUNCTION GET_AGE_CATEGORY (

 P_EID IN NUMBER

)

RETURN VARCHAR2

IS

 V_AGE EMP.AGE%TYPE;

 V_CATEGORY VARCHAR2(20);

BEGIN

 SELECT AGE

 INTO V_AGE

 FROM EMP

 WHERE EID = P_EID;

 IF V_AGE < 18 THEN

 V_CATEGORY := 'Minor';

 ELSIF V_AGE <= 60 THEN

 V_CATEGORY := 'Adult';

 ELSE

 V_CATEGORY := 'Senior';

 END IF;

 RETURN 'Age: ' || V_AGE || ', Category: ' || V_CATEGORY;

EXCEPTION

 WHEN NO_DATA_FOUND THEN

 RETURN 'Employee not found';

END;

/

SET SERVEROUTPUT OFF

SET VERIFY ON

SET FEEDBACK ON

Output:

15. Write a PACKAGE which includes Package Declaration, Package Body and call the individual object from the Package.

```
/*
Write a PACKAGE which includes Package Declaration, Package Body and call the
individual object from the Package.
*/

SET SERVEROUTPUT ON
SET VERIFY OFF
SET FEEDBACK OFF

CREATE OR REPLACE PACKAGE my_package
AS
    PROCEDURE greet;
END my_package;
/

CREATE OR REPLACE PACKAGE BODY my_package
AS
    PROCEDURE greet
    IS
    BEGIN
        DBMS_OUTPUT.PUT_LINE('Hello World');
    END greet;
END my_package;
/

BEGIN
    my_package.greet;
END;
/

SET SERVEROUTPUT OFF
SET VERIFY ON
SET FEEDBACK ON
```

Output: